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CAS-Ohm 1.0: A Profile-Specific Experimental Response and Calibration Model

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Abstract

CAS-Ohm 1.0 is a typed proposal for modeling a local, profile-specific response between a declared intervention coordinate (steering alpha) and an observed activation coordinate (activation projection) in a neural system under audit. It is *not* physical Ohm's law, an electrical implementation, a measure of attention, or a causal law. Its candidate prediction, `y_hat = a / r_eff`, is admissible only when the effective resistance `r_eff` carries model-specific units of alpha per activation, was fit on a registered calibration split, carries uncertainty, and is evaluated on a disjoint held-out split; defining resistance

as alpha over response on the same samples and reporting the identity as confirmation is rejected as circular.

The contribution is threefold. First, a quantity-and-evidence ontology in which every numeric value is typed evidence — quantity kind, exact decimal, unit, measurement kind, method, calibration-profile digest, uncertainty, lineage, and censoring status — and physical volt/amp/ohm labels are forbidden aliases for model quantities. Second, a source audit that preserves eight conflicts or under-specifications across the three CAS-Ohm sources (incompatible scales, incompatible full-scale methods, a missing dimensional bridge, distinct gain definitions, overloaded ASOM axes, unvalidated network rules, non-identifying residuals, under-specified quantization boundaries) rather than harmonizing them silently. Third, a fail-closed software candidate implementing exact Q4 classification, structural-only series/parallel/gain network calculations, non-identifying residual diagnostics, and a strict separation between structural research-packet completeness and authenticated Evidence Packet eligibility — the latter always false in this candidate. At the bound Gate 4 receipt, 65 CAS-Ohm test methods and 58 parameterized subtests pass; these are bounded software tests, not live-model experiments. Five registered empirical hypotheses await preregistered closure through the Gate 5 control kernel.

---

1. Scope and authority boundary

CAS-Ohm borrows the *shape* of Ohm's law — a dose coordinate, a response coordinate, and a fitted proportionality parameter — as an experimental hypothesis about local linear response regions in steered model activations. The borrowed shape licenses nothing by itself: dimensional consistency is necessary for a model relation [E12] but not sufficient for empirical or causal validity. This report establishes no live execution, scientific conclusion, physical measurement, causal effect, safety or clinical conclusion, product capability, license, canon admission, or disclosure.

2. Preserved source conflicts

The audit records eight conflicts across the Guide, the operationalization addendum, and PCBA Appendix I; none is silently harmonized.

Conflict	Finding	Disposition
Scale and saturation	Guide `R` range `0.1–10` vs Appendix I dimensionless `0–100` score; `sat=8.5` vs a `0–1` region fraction	Q10, Q100, fractions, alpha, activation, and effective resistance kept as separate types
Full-scale profile	Board-selected, per-path rail/floor, and corpus 95th-percentile methods differ	Separate named profile methods with digests
Dimensional bridge	`I_pred = V/R_total` invalid until resistance has calibrated `alpha_unit / activation_unit`	Require a calibrated bridge; scores cannot substitute

Conflict	Finding	Disposition
Gain definition	Fisher-Rao cause/effect gain vs local Jacobian/projection gain	No identification without a proved coordinate bridge or experiment
ASOM axes	Dose-response region and semantic correctness overloaded	Classified independently (§4)
Network rules	Series/parallel/gain presented more strongly than evidence permits	Capability-gated structural calculations only (§6)
Residual identification	Residual sign read as hidden policy/bypass/cause	Non-identifying diagnostic with uncertainty and censoring (§7)
Q4 boundaries	Two-decimal display bands ambiguous at high precision	Exact half-open bins at 2.5, 5, 7.5 (§5)

Preserving this record is itself an engineering result: downstream code cannot claim one coherent inherited scale or gain law where the sources do not provide one.

3. Quantity and evidence ontology

Every numeric value is typed evidence, never a bare binary float:

```
ObservedValue {
  quantity_kind, exact_decimal_value, unit_id,
  measurement_kind, method_id, calibration_profile_sha256,
  estimate_or_observation, uncertainty, confidence_interval,
  sample_size, raw_artifact_refs, control_refs,
  derivation_parents, missing_or_censored_reason
}
```

Measurement kinds — `DESIGN_VALUE`, `INSTRUMENTED`, `BEHAVIORAL_PROXY`, `MODEL_DERIVED`, `HUMAN_SCORE`, `LLM_SCORE` — are not fungible: a derived value does not become instrumented because one parent was, and a score does not become telemetry because it is stored next to a measurement. The unit ontology separates constraint score, normalized fraction, alpha dose, activation projection, effective resistance, and real physical SI units; physical volt/amp/ohm labels are forbidden aliases. This follows basic metrological discipline (JCGM vocabulary [E6]; uncertainty guide [E7]) without claiming that CAS-Ohm *is* metrology.

Canonical evidence is exact-type tagged, cycle-aware, unsafe-Unicode rejecting, and bounded by declared depth, node, container, text, output, and 2,048-bit integer limits; host-runtime conversion failures normalize to the candidate's validation error rather than leaking platform exceptions.

4. Calibration and independent classifications

A calibration profile binds exact model/config/tokenizer/weights and code identities; runtime and dtype; test-point and direction-vector identity, norm, and sign; corpus manifest and rights; seeds; alpha grid; full-scale method; fit and uncertainty methods; tolerances; response-region rules; controls; drift/expiry; review state; and a canonical digest. Effective resistance is estimated on calibration data

and must predict held-out response; calibration must represent parameter uncertainty and model discrepancy rather than treating the best fit as truth [E15]. Verification of the software and operational validation of the response model remain distinct activities [E8].

Two axes are classified independently:

- **Dose-response region:** `CUTOFF`, `ACTIVE`, `SATURATION`, `NON_OHMIC`, or `UNSTABLE`.
- **Semantic outcome:** `INTENDED`, `OPPOSITE`, `OFF_AXIS`, or `UNASSESSED`.

No region mechanically determines an outcome — an active response may be off-axis or opposite; a flat response may be cutoff, saturation, a wrong direction, insensitivity, or measurement failure. ASOM is retained as a mnemonic, not an identity theorem. No diagnosis is emitted from a cutoff, saturated, unstable, uncalibrated, stale, or wrong-model profile.

5. Exact normalization and Q4 quantization

For profile-specific full-scale activation `I_FS`, the classifier computes the exact pre-clipping value `exact_q10 = 10 · raw / I_FS`, retains `overload_ratio = raw / I_FS` separately, and clips only the displayed `q10`. Exact-input bands use half-open bins:

```
[0, 2.5) -> Q1     [2.5, 5) -> Q2     [5, 7.5) -> Q3     [7.5, ∞) -> Q4
```

Display rounding cannot move a value across a band boundary. Trend tags require an explicit axis, interval, minimum effect, and uncertainty rule. The choice of `I_FS` is part of the calibration profile; the three inherited full-scale methods are not interchangeable.

6. Structural network operations

Series addition, reciprocal parallel aggregation, and gain-product composition exist only as unauthenticated structural research calculations, each capability-gated: series requires a simple acyclic chain, compatible coordinates, one unit, active-region components, adequate linearity, one profile, and complete ordered child provenance; parallel requires genuine alternative directed branches with shared endpoints (co-present guardrails are not automatically parallel paths); gain composition requires adjacent coordinate agreement, dimensionless local gains, and active-region status — and Fisher-Rao displacement gain is not automatically the local Jacobian/projection derivative. Every aggregate binds exact child order, topology, profile identity, evidence, runtime/code/dependency identities, covariance limitations, and trust mode; duplicate or cyclic paths fail closed. Optional Ed25519 fixture-key attestations are integrity/replay checks only — never independent reviewer, legal, product, or Evidence Packet authority.

7. Residuals are non-identifying

For an admitted active-region prediction, `delta_y = y_observed − y_predicted` requires instrumented held-out observations, matching test points, profile identity, a registered prediction, calibrated resistance, and propagated uncertainty; at zero or zero-compatible predicted response the relevant ratios are censored or undefined. A residual can rank follow-up probes. It cannot, by sign alone, identify a hidden policy, bypass, opposition, cause, or safety failure — causal interpretation requires an explicit causal model and identifiable interventions [E11]. The registered residual-localization claim therefore calls for blinded implanted faults with matched controls and remains untested.

8. Research-packet completeness versus authenticated eligibility

Structural research-packet completeness requires an exact model revision, valid profile digest, raw artifact bytes, relevant controls, uncertainty, per-field lineage, a nonstale reviewed calibration profile, and at least one genuinely instrumented field; score-only, zero-instrumentation, mixed-scale, unknown-profile, or tampered records fail closed. Authenticated Evidence Packet eligibility is a separate axis and is *always false* in this candidate: it requires an independently admitted authority manifest, protected key lifecycle and revocation, trusted current-time receipt, materialized model/code/runtime dependency identities, authenticated instrument evidence, and a reviewed packet policy. Deterministic in-repository fixture keys and caller-supplied replay time never satisfy those conditions.

9. Bounded software evidence and registered hypotheses

At the bound Gate 4 receipt the CAS-Ohm suite records 65 passing test methods (test-ID SHA-256 `2E81A685961286D944894A22EB93345B97211B4433B6C71EAF628D84CD8EB1EF`), 58 passing parameterized subtests, eight retained source conflicts, and zero authenticated Evidence Packet eligibility paths. The suite exercises quantity/unit rejection, canonical evidence limits, calibration identity, split separation, exact Q4 boundaries, structural network constraints, residual censoring, uncertainty, lineage, and fail-closed capability behavior. Synthetic known-fault boards (linear, cutoff, saturation, nonlinear, parallel, confounded, bypass) validate software behavior only.

Five registered hypotheses, all `IN_REVIEW / Ring 4 / NOT LICENSED / HELD` with no admitted Evidence Packet:

1. `CLM-CAS-OHM-LOCAL-RESPONSE` — held-out local response to randomized alpha;
2. `CLM-CAS-OHM-EFFECTIVE-RESISTANCE` — held-out prediction plus independent constraint manipulation;
3. `CLM-CAS-OHM-GAIN-COMPOSITION` — local-to-end-to-end derivative composition;
4. `CLM-CAS-OHM-REDUNDANCY` — genuine-branch ablation over matched random-head controls;
5. `CLM-CAS-OHM-RESIDUAL-LOCALIZATION` — blinded implanted-fault localization above shuffled and matched-control baselines.

`CLM-CAS-OHM-PHYSICAL-TELEMETRY` is recorded as a `NOT LICENSED` product claim; current evidence contradicts any representation of design scores, behavioral proxies, or model-derived values as physical telemetry.

10. Preregistered closure path

A future run must use the Gate 5 per-claim preregistration and Evidence Packet v2 control kernel: freeze exact claim text, model/checkpoint/tokenizer/weights, code, environment, test point, direction, corpus, splits, seeds, alpha grid, fit and uncertainty methods, controls, falsifiers, and permitted claim language; keep calibration and validation splits disjoint; include zero-intervention, random same-norm direction, wrong-layer, wrong-token, label-shuffle, and norm-matched controls; record cutoff, saturation, instability, drift, missingness, exclusions, and negative outcomes without erasure; require independent numerical recomputation and authenticated execution evidence; replicate on a held-out checkpoint or model under a prespecified comparison rule; and keep scientific eligibility `NOT_ESTABLISHED` until independent audit, replication, claim review, and external scientific review exist.

11. Memristive extension boundary

Classical memristive-system theory [E13, E14] motivates a mathematical question — whether a CAS response parameter should depend on a versioned history state — without authorizing a physical mapping. A provisional nonphysical extension would declare

```
x_(t+1) = F_profile(x_t, a_t, y_t, context_t; θ)
r_eff,t = G_profile(x_t; θ)
```

with `x_t` an explicit model-state record (initialization, update law, sampling interval, reset rule, drift model, uncertainty, provenance), typed separately from both a memoryless fit and an electrical memristor, and required to predict held-out hysteresis beyond a matched memoryless baseline. Absent that evidence, "memristive" remains an analogy and research hypothesis. No stateful implementation, calibrated history variable, hysteresis experiment, material device, or safety property is admitted.

12. Limitations

Inherited scale conflicts are preserved, not resolved. There is no admitted live calibration, held-out estimate, drift-threshold validation, or inter-rater reliability result. The implemented prediction is a guarded software contract, not a validated law. Network rules are structural-only. Gain, redundancy, and residuals remain noncausal. History-dependent dynamics remain open. No CAS quantity is a physical electrical quantity. No safety, clinical, regulatory, product, canon, license, or disclosure authority exists. Internal cross-review and independent external review have not occurred.

13. Conclusion

CAS-Ohm 1.0 is viable as a profile-specific experimental response and calibration proposal precisely because its type system makes the Ohm analogy *testable and rejectable*. Its strongest present achievement is not a discovered cognitive law but the enforced separation of quantities, units, measurement kinds, profiles, uncertainty, response regions, semantic outcomes, network structure, and authority — a design in which the circular validation, the unit pun, and the residual-as-cause inference are each unrepresentable or fail closed. Whether the model predicts anything is a question the five registered hypotheses leave to preregistered, controlled, independently audited experiments.

References

Identifiers `[E5]–[E15]` resolve to complete metadata in `WAVE1_ENGINEERING_CITATION_LEDGER_v0.1.json`.

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